Post-Quantum Broadcast Authentication in Per-Message Byte Budgets: Fit and Measured Behaviour of Ten Modes Across Channels

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Timing numbers are from a single cloud VM (`cloud-vm-4core`, 2026-11-13, commit `694ba76`, clean tree) pending runs on the authors' machines. Every value is defined in `numbers.tex` and traced to `results/summary.md`; open items are marked TODO-RESULT or [TODO].

***Abstract*—Handshake-free broadcast channels (drone Remote ID, 5G SIB1, Galileo OSNMA, C-V2X) give each message a fixed number of bytes for authentication. We derive these budgets from their sources, separating the payload ceiling from the authentication budget, and score 24 lowest-level liboqs size classes (64 overall) by declared size: a full post-quantum (PQ) signature fits one message for 5 to 11 of 24 classes, depending on the profile. We then implement 10 authentication modes with liboqs (fragmented and erasure-coded signatures, hybrids, Merkle-batch signing, and three variants of a PQ-anchored TESLA-style chain), check them with 427 negative tests, and compare them at an equal channel budget on SIB1 and two C-V2X profiles under synthetic i.i.d. and burst loss. Against erasure-coded per-message signatures, the chain authenticates a larger share of messages (supported: 118 wins, 3 losses, 341 ties; 22 of the wins are cells where fragmentation cannot run). It is refuted on bytes on air per authenticated message (131/290/19), authentication delay (0/440/0) and late-joiner time (25/385/30). The chain also gives up non-repudiation. All timings come from one x86 cloud VM.**

***Index Terms*—post-quantum signatures, broadcast authentication, TESLA, V2X, Remote ID, GNSS, 5G**

I. Introduction

Several cyber-physical broadcast channels carry messages that a passive receiver must authenticate without a handshake: drone Remote ID over Bluetooth pages [1], 5G SIB1 [2], Galileo OSNMA [3] and C-V2X basic safety messages (BSMs) [4], [5]. Each has a hard byte limit per message. PQ signatures, whether standardized (ML-DSA [6], SLH-DSA [7], LMS/XMSS [8]) or selected (Falcon as FN-DSA [TODO: cite: FN-DSA standard, none published yet]), are larger than the classical signatures they replace.

Prior work answers the fit question per domain [2]–[5], [9] or across algorithms within one domain [10], [11]. Budget definitions differ between these studies, and so do the conclusions: Falcon-512 is "insufficient" for C-V2X in [4] and "feasible" in [5]. Amortizing modes, such as a TESLA-style chain anchored by one PQ signature per epoch [2], have so far been evaluated within one domain. This paper measures fit and loss behaviour on common terms. It proposes no new mechanism and gives no security proof.

Contributions.

- *Budget profiles* for five channels, taken from their sources, that separate the payload ceiling from the authentication budget and mark every derived value (Table I).
- *Fit* of 24 lowest-level size classes (64 overall) by declared signature size, with PQ-issuer certificates, key delivery, and a two-ceiling explanation of the Falcon-512 C-V2X conflict (Sec. IV-A).
- *Real implementations* of 10 authentication modes on liboqs, verified by 427 negative tests, and compared at an equal channel budget on three profiles (Sec. IV-B).
- *The measured finding*: the chain authenticates a larger share under loss, but is refuted on bytes per authenticated message, authentication delay and late-joiner time (Table III).

II. Related Work

Per-domain fit. Twardokus et al. [4] argue that the C-V2X payload cannot carry a PQ signature and propose a partially hybrid V2V protocol; Abrar et al. [5] find Falcon-512 feasible within an SAE J3161 transport block. Chen et al. [12], [13] study hybrid PQ/ECC certificates for the V2X credential system [14]; Eaton et al. [15] give PQ butterfly key expansion (abstract only). Junquera-Sánchez et al. [3] assess PQ options for OSNMA. Darzi et al. [2] anchor an HMAC chain with one PQ signature per epoch to fit 5G SIB1. Goertzen and Stebila [9] fragment PQ DNSSEC responses; DNS is request–response, so we keep it only as a ceiling reference. TBRD [16] uses TESLA for Remote ID, and Hussain et al. [17] a fast MAC for grid sampled values.

Multi-algorithm, one domain. Yoshizawa et al. [10] compare NIST and additional-round signatures for V2X and compress them with zero-knowledge proofs; Mamun et al. [11] review PQ cryptography for ITS; Nabasirye et al. [18] describe a benchmarking framework on Xtensa hardware (abstract only).

Broadcast authentication and amortization. Classical work signs multicast streams over lossy channels [19], amor-

TABLE I
BUDGET PROFILES (BYTES). *DERIVED FROM STATED NUMBERS. †CEILING-BASED: NO AUTHENTICATION-ONLY BUDGET IN THE SOURCE. JOIN DEADLINES ARE OUR ASSUMPTIONS.

Profile	Ceiling	Auth	Period	Join dl.	Src.
Remote ID	255	201	3 s	30 s	[1]
5G SIB1	372	293	0.16 s	10 s	[2]
OSNMA KROOT (sig.)	208	158*	480 s	720 s	[3]
OSNMA PKR (key)	208	76*	480 s	–	[3]
C-V2X 437†	437	374*	0.1 s	10 s	[4]
C-V2X J3161†	2481	2414*	0.1 s	10 s	[5]
DNS†	1232	1232	TODO-RESULT	1 s	[9]

tizes signatures over packets [20], signs flows and multicasts [21], and builds short one-time signatures [22]; we read only titles or abstracts of these, so we do not claim how closely our Merkle and erasure-coded modes match their constructions [TODO: check full text: wong1998digital, park2002efficient]. TESLA [23] authenticates by delayed key disclosure under loose time synchronization, without non-repudiation. LMS [24] and XMSS [25] bound signatures per key; few-time schemes [26], [27] trade capacity for speed.

Hybrids, MCUs, envelopes. RFC 9955 [28] defines hybrid signature properties; Fall [29] surveys hybrid deployment and Devevey et al. [30] build non-separable hybrids (both abstract only). Cortex-M results exist for ML-DSA [31], MAYO [32], [33], SQIsign [34] and Falcon [35], [36], and pqm4 covers more [37]; all but [31] were read as abstracts. Hyoung et al. [38] build a size–latency envelope for a ROS2/DDS handshake (abstract only), and Sikeridis et al. [39] study PQ signatures in TLS 1.3. We cannot exclude an existing broadcast-focused systematization [TODO: check full text of hyoung2026post and fall2025sok]. UOV, MAYO, SNOVA, CROSS and MQOM are candidates in NIST's additional-signature process, not standards [TODO: cite: NIST additional signatures round status].

III. METHOD

A. Budget profiles

Table I lists each profile. Remote ID allows 201 B of authentication data (RFC 9575 caps the 255 B field) [1]; we use its 3 s static-message period, and its 1 s dynamic period would shrink every message-wait term proportionally. SIB1 has 293 B after the 79 B of SIB1 content in [2]. OSNMA's block arithmetic is internally inconsistent in the source [3]; we score signatures only against the DSM-KROOT signature space (158 B, from 79 bit per block) and public keys only against the DSM-PKR key space (76 B left after the Merkle proof). Frame counts use the bit capacity. C-V2X and DNS state no authentication-only budget, so we subtract stated overheads from the ceiling with one rule for both C-V2X ceilings: SPDU header (24 B [4], 28 B [5]) and the 39 B BSM core of [5], borrowed for the 437 B case (TODO-RESULT). If the 40 B frame overhead lies inside 437 B, the budget drops to 334 B; the fit count is 8 at both values. Verdicts on these profiles are conditional on the derivation.

B. Scheme set and fit accounting (Q1)

liboqs 0.16.0 [40] enables 221 signature parameter sets, which collapse to 64 classes of distinct (family, level, key size, signature size); main tables use the 24 lowest-level classes. S is the *declared* maximum signature size (Falcon-512: 752 B). 5 LMS/XMSS rows enter as literature sizes. A mode fits if its per-message object $x \le B$; otherwise it needs $\lceil x/B \rceil$ messages. Modes: full ($x = S$); hybrid (S plus the source's classical signature); a certificate with an explicit *PQ issuer* of the same scheme (peak $x = 2S + P +$ metadata) where the source sends certificates (C-V2X); the source's *classical-issuer* certificate ($P+105$ B, derived from [5]) shown separately; signature plus bare key where the source sends one (Remote ID, DNS); and key delivery ($x = P$, OSNMA-PKR). Certificate metadata is unknown (TODO-RESULT); we use 0 B as a lower bound and report 41 and 73 B as sensitivity values. The chain always fits per message, so it is judged by its join time against the deadlines of Table I.

C. Protocol implementations (Q2)

Sender and receiver run real cryptography and byte framing: liboqs through a C shim, HMAC-SHA-256 and a SHA-256 chain truncated to 16 B, and systematic Reed–Solomon codes (zfec 1.6.0.0). A message counts as authenticated only if the receiver verified the received bytes. The 10 modes are: frag_rep/frag_mds, the signature fragmented and repeated or MDS-coded within the message's window; hyb_rep/hyb_mds, the same with Ed25519 plus the PQ signature, both verified; merkle4/8/16, one signature per batch of n messages with a $\log_2 n \times 32$ B path per message, the signature MDS-coded over the next batch's windows; chain, a 16 B HMAC tag per message, the key disclosed one window later, and an anchor (16 B commitment, 4 B timestamp, PQ signature) MDS-coded across the epoch's slots, with the wait for the anchor counted in the delay; chain_lead, which sends epoch e's anchor during epoch $e-1$; and chain_lead_cont, which also carries the next commitment inside the MAC input so that an authenticated epoch needs no new anchor. Receivers enforce the TESLA time-safety check [23] with an exactly known schedule. Public keys are pre-provisioned.

Equal channel budget. One allocation rule applies to all modes. Every application message gets M slots ($M \in \{10, 40\}$) of the same capacity (293 B on SIB1, 366 B and 2406 B on the two C-V2X profiles after the 8 B digest). Every usable slot carries one frame of header, mode control field and one symbol of the mode's heavy object (signature, batch signature or anchor). On C-V2X the slot carrying the full certificate (every 5th) usually has no room left and carries no authentication data. A mode is infeasible if its span has fewer usable slots than the object has source symbols. Our frames carry their own explicit headers (3–11 B plus control fields); the sources' fragment headers are unknown (TODO-RESULT).

Channel and metrics. Each run sends 80 messages; epochs are 20 windows. Loss is synthetic and per slot: i.i.d. at $p \in \{0.02, 0.05, 0.1, 0.2, 0.3\}$ and Gilbert–Elliott with mean bursts

TABLE II
LOWEST-LEVEL CLASSES FITTING ONE MESSAGE (OF 24). [a] CERTIFICATE METADATA 0/41/73 B. LAST ROW: CHAIN JOIN TIME (NO LOSS) WITHIN THE DEADLINE OF TABLE I. † CEILING-BASED.

Mode	RID	SIB1	KROOT	CV437†	J3161†	DNS†
full	7	8	5	8	11	11
hybrid	5	7	0	8	11	11
cert., PQ issuer[a]	–	–	–	0	4/3/3	–
cert., classical issuer	–	–	–	0	4	–
signature + bare key	0	–	–	–	–	0
chain join $\leq$ deadline	11	23	0	24	24	n/a

TABLE III
CHAIN VARIANTS VS. FRAG_MDS: CELLS WON/LOST/TIED OVER ALL PROFILES, SCHEMES, M AND LOSS CONDITIONS (RULE IN SEC. III). VERDICT IS THE SAME FOR ALL THREE VARIANTS.

Hypothesis	cells	chain	lead	lead+cont	verdict
H2a share ↑	462	118/3/341	118/3/341	118/3/341	supported
H2b B/auth. msg ↓	440	131/290/19	124/309/7	100/324/16	refuted
H2c delay ↓	440	0/440/0	0/440/0	0/440/0	refuted
H2d join ↓	440	25/385/30	0/439/1	0/439/1	refuted

of 3, 5, 10 slots at stationary loss 0.1, 0.2, 0.3. We use 20 seeds with the same loss mask for every mode (100 800 rows over 7 PQ schemes). Metrics are the share of messages authenticated, bytes on air per authenticated message, median authentication delay, and the late joiner's time to first authentication (joining at a random slot of the first epoch).

Hypotheses and decision rule (fixed before the data). The chain should beat frag_mds on share (H2a), bytes per authenticated message (H2b), delay (H2c) and join time (H2d). Per cell (profile, scheme, M, loss condition) we take the paired per-seed difference. The cell is a win if the mean is favourable, exceeds 3 standard errors and a minimum effect (share 0.005, otherwise 1%), a loss if the same holds in the other direction, and a tie otherwise. An infeasible mode has share 0; the other metrics are compared only where both modes authenticate in every seed. A hypothesis is supported if wins $\geq 2\times$ losses (and wins > 0), refuted if losses $\geq 2\times$ wins (and losses > 0), and inconclusive otherwise.

Implementation checks. 427 of 427 negative tests pass over 7 PQ schemes and all modes: tampered payload, tags, paths, signature symbols and disclosed keys; wrong signer; forgery with a disclosed key; replayed frames; corrupted anchors; forged next-epoch commitments; and hybrids with an invalid Ed25519 half. A receiver without the time-safety check accepts 21 of 21 late-key forgeries, which shows that the check is what stops them. Closed forms agree with the measured share in all 1716 comparable i.i.d. cells (0 beyond 3 standard errors, maximum difference 0.0175). Since the formulas share the model's assumptions, this is a code check, not a validation. Sender and receiver as separate processes over UDP loopback reproduce the in-process outcome in 4 of 4 runs.

IV. RESULTS

A. Q1: what fits a single message

A full signature fits one message for 5 to 11 of 24 classes (Table II, Fig. 1). Under the authentication budgets (Remote ID, SIB1, OSNMA) every fitting class is UOV, SNOVA or MAYO; Falcon fits only the J3161 and DNS ceilings. ML-DSA-44 (2420 B) fits nothing; its J3161 margin is -6 B, so that verdict rests on the derivation. CROSS, MQOM and SLH-DSA fit no profile. When the signature does not fit, ML-DSA-44 needs 9 SIB1 messages (1.28 s without loss) and 13 Remote ID messages (36 s); on OSNMA, Falcon-512 needs 5 messages (2.28×10^3 s). A signature plus bare key fits no class on Remote ID or DNS. On the OSNMA key space, 9 of 24 public keys fit, all SLH-DSA, CROSS or MQOM, the families whose signatures fit nowhere. The chain's join time meets its deadline for all classes on both C-V2X profiles, for 11 on Remote ID (up to 276 s), and for none on OSNMA (810 s to 5.652×10^4 s). Among stateful rows only LMS with 32 signatures per key (3.2 s of 10 Hz BSMs) fits J3161, which is operationally empty; larger trees are TODO-RESULT.

Certificates. With a PQ issuer, 4 classes fit J3161 at zero metadata and 3 at 41 or 73 B; none fits C-V2X 437. The source's classical-issuer certificate also gives 4 on J3161, but it is not a PQ chain.

The Falcon-512 C-V2X conflict. Twardokus et al. use the 437 B maximum payload of a 10 MHz channel [4]; Abrar et al. assume a J3161 20 MHz profile at MCS 11 with 10 subchannels, a 2481 B transport block [5]. Under our single subtraction rule, Falcon-512 (752 B declared) needs 3 messages under the first ceiling (margin -378 B) and fits the second with 1662 B to spare. With a Falcon-512 issuer the certificate message fits by only 13 B at zero metadata, so it stops fitting at 41 B; with the source's classical issuer the margin is 660 B. Both papers are consistent with their own ceilings. "Feasible" in [5] means fitting a transport block; their own 90% delivery target holds only in the lightest line-of-sight scenario.

Consistency checks. Dividing the sources' own crypto-byte totals by 293 B gives the 13 and 34 SIB1 packets of [2]; the source's 79 bit per block gives 68 OSNMA blocks where [3] reports 67. These check arithmetic and are not reproductions.

B. Q2: measured comparison at equal channel budget

Table III gives the four verdicts and Table IV sample cells.

H2a, share: supported, with limits. The chain wins 118 of 462 cells, loses 3 (all Falcon-512 on J3161) and ties 341. Of the wins, 22 are cells where frag_mds cannot run at all (SLH-DSA-128s at $M = 10$ on SIB1 and C-V2X 437) and 96 are cells where it runs and loses. The largest gaps are for signatures that need many slots, e.g. ML-DSA-44 on SIB1 at $p = 0.1$ (0.721 vs. 1.000) and on C-V2X 437 at $p = 0.3$ (0.247 vs. 1.000). Under heavy bursts (stationary loss 0.3, mean burst 10 slots, SIB1) the gap is smaller for Falcon-512 (0.830 vs. 0.888), and OV-Is, whose signature fits one slot, reaches 0.892. On J3161, where every signature but SLH-DSA fits one or two slots, the chain wins only 16 cells. The chain reaches share ≥ 0.95 at $M = 10$ for all 21 (profile, scheme) pairs; frag_mds does so for 17 and needs $M = 40$ for the other 4.

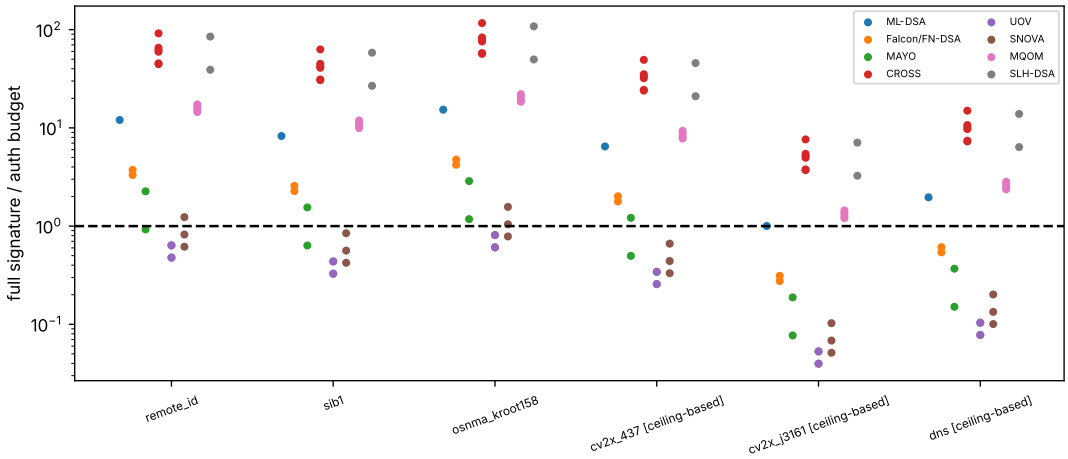


Fig. 1. Declared full-signature size over authentication budget, one dot per lowest-level liboqs class; below the dashed line fits one message.

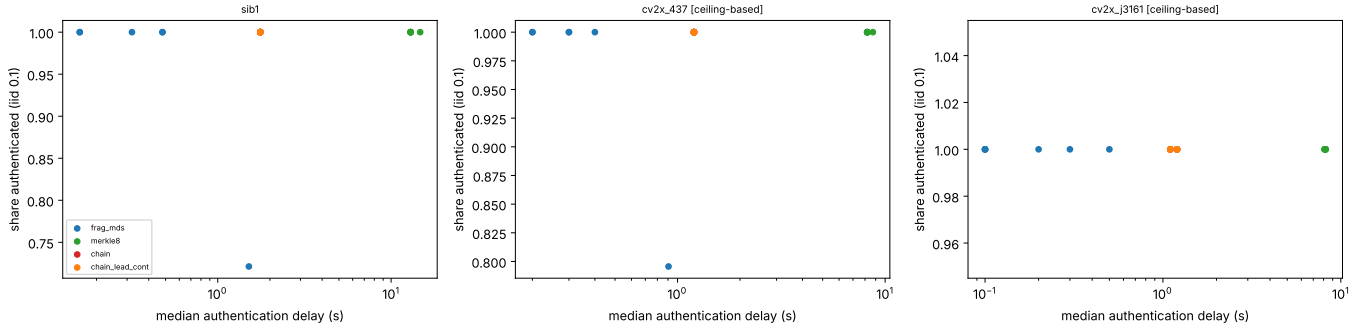


Fig. 2. Share authenticated against median authentication delay (log scale) at i.i.d. loss 0.1, $M = 10$, one point per PQ scheme. chain points lie under chain_lead_cont (equal delay and share).

TABLE IV
SIB1, I.I.D. LOSS 0.1, $M = 10$ (2930 B ALLOCATED PER MESSAGE): SHARE AUTHENTICATED, MEDIAN DELAY AND JOIN TIME (MS), BYTES ON AIR PER AUTHENTICATED MESSAGE. MEANS OVER 20 SEEDS.

PQ	mode	share	delay	join	B/auth
ML-DSA-44	frag_mds	0.721	1512	2872	3807
	merkle4	1.000	6624	6576	2910
	merkle16	1.000	25760	25480	2860
	chain	1.000	1760	1856	2885
	chain_lead_cont	1.000	1760	17376	2965
Falcon-512	frag_mds	1.000	480	584	2550
	merkle4	1.000	6560	6080	2590
	chain	1.000	1760	1080	2375
	chain_lead_cont	1.000	1760	17376	2685
OV-Is	frag_mds	1.000	160	176	1010
	merkle4	1.000	6560	5840	1680
	chain	1.000	1760	1032	1615

H2b, bytes per authenticated message: refuted (131/290/19). Because every usable slot carries a full frame in every mode, bytes on air per message are nearly equal across modes, and the metric mostly reflects the share in its denominator. At $p = 0.1$, $M = 10$ the chain sends a median 1.132 times the bytes of frag_mds per authenticated message (range 0.706–1.599 over 19 pairs). The chain's small per-message tag therefore does not translate into fewer bytes under an equal allocation; whether a smaller allocation would suffice is answered only indirectly by the target-share counts above.

H2c, delay: refuted in every cell (0/440/0). The chain must wait one window for key disclosure (1600 ms on SIB1, 1000 ms on C-V2X at $M = 10$) and for the anchor; a signature that fits a few slots authenticates within its own window (Fig. 2). Merkle batching is slower still and grows with n: for Falcon-512 on SIB1 the median delay is 6560 ms at $n = 4$ and 25760 ms at $n = 16$, both at full share, so we report all $n \in \{4, 8, 16\}$ rather than one.

H2d, late-joiner time: refuted (25/385/30). A joiner must collect an anchor before it can authenticate anything. The chain wins some cells where fragmentation itself joins slowly, e.g. ML-DSA-44 on SIB1 (Table IV). Sending the anchor one epoch ahead (chain_lead, chain_lead_cont) removes the steady-state anchor wait, but the one-window disclosure delay remains (H2c unchanged), and join time is lost in all but one cell.

Bursts. At stationary loss 0.3 with mean bursts of 10 slots, the share changes relative to i.i.d. loss by -0.119 for the chain, -0.076 for frag_mds and -0.223 for merkle8 (means over profiles and schemes); the effect is not uniform: ML-DSA-44 frag_mds on SIB1 rises from 0.142 to 0.539.

SLH-DSA-128s. Per-message signing is infeasible in 8 of 24 (profile, M, fragment/hybrid mode) configurations, and its signing time (median 591 ms) exceeds one slot in 24 of 24. Across all 202 feasible (profile, scheme, mode) configurations at $M = 10$, 0 senders are slower than their message rate. SLH-DSA is usable only where a batch or anchor amortizes it or the budget is large (J3161).

C. Q3: size or timing

With timing-bound meaning that the p99 sign or verify time exceeds the packet period, size binds in 7 of 8 families and timing in 0 (UOV not distinguishing). This is close to tautological: packet periods are long against x86 operation times (median sign 235 μs for Falcon-512, 68 μs for ML-DSA-44), and it says nothing about microcontrollers.

V. Security Properties and Limitations

Properties (stated, not proved). The chain gives no non-repudiation: any receiver holding a disclosed key can compute valid tags. It cannot replace signed messages where misbehaviour reporting or revocation needs attributable messages, as for V2X BSMs [14]; only the per-message and Merkle-batch modes give transferable evidence. The chain needs loose time synchronization; our runs assume zero clock error, and a manipulable receiver clock is an attack surface. Chain continuity (chain_lead_cont) survives a lost anchor but makes later epochs depend on earlier MACs. Hybrids require both signatures to verify; downgrade resistance beyond that is not analysed [28]. The negative tests check the implementation, not the protocols' security.

Limitations.

- *One machine.* All results come from one 4-core cloud VM (`cloud-vm-4core`, CPU TODO-RESULT, commit `694ba76`). No MCU timings were measured [TODO: cite or measure: Cortex-M sign/verify for UOV, SNOVA, MAYO]; Q3 is x86 only.
- *Synthetic channel.* Loss is i.i.d. or Gilbert–Elliott per slot. Slots are collision-free: there is no radio scheduling (e.g. C-V2X semi-persistent scheduling), no congestion and no measured traces, so the C-V2X results are not packet-delivery predictions. The UDP mode was run on loopback only, not across two machines.
- *Assumed parameters.* Tag, key, commitment and timestamp sizes, disclosure delay, epoch length, M and join deadlines are assumptions. Q2 covers SIB1 and the two C-V2X profiles, not Remote ID, OSNMA or DNS.
- *Budgets and certificates.* C-V2X and DNS budgets are ceiling-based and some verdicts sit within a few bytes of them; OSNMA budgets come from inconsistent source arithmetic; DNS has no message rate (TODO-RESULT). Certificate metadata is unknown (0 B used as lower bound); intermediate CAs and mixed PQ-CA/classical-subscriber chains [13] are not modelled. Public keys are pre-provisioned except in the key-delivery and certificate rows.
- *No reproductions.* Not reproduced: the OTA overhead (TODO-RESULT) and 10%-loss point (TODO-RESULT) of [2], the key-block figure (TODO-RESULT) of [3], the transfer sizes of [9] (TODO-RESULT), the ZK compression of [10] (TODO-RESULT) and the tables of [12] (TODO-RESULT).

VI. Conclusion

Under budgets taken from each channel's sources, only UOV, SNOVA and MAYO carry a full PQ signature in one Remote ID, SIB1 or OSNMA message, and Falcon-512 fits C-V2X only under the larger J3161 ceiling, where a PQ-issued certificate leaves almost no margin. Measured with real implementations at an equal channel budget, a PQ-anchored chain authenticates a larger share of messages than erasure-coded per-message signatures, mainly where the signature needs many slots; bursts hurt it more than them on average. It does not use fewer bytes per authenticated message, it authenticates later in every cell, it is slower for late joiners, and it gives up non-repudiation. Next steps are runs on the authors' machines, MCU timings, measured loss traces and a scheduled C-V2X channel.

Artifact

Scripts that regenerate every number from raw CSVs: [TODO: artifact URL].

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